Contents

Chapter 1

QUBIC: A Bolometric Interferometer (BI)

Contents

Introduction

Observations of the Cosmic Microwave Background (CMB) have become one of the most powerful probes in modern cosmology, providing stringent constraints on the composition, geometry, and evolution of the Universe. In particular, the polarisation of the CMB carries unique information about the physics of the early Universe. Of special interest is the curl-like component of the polarisation field, known as B-modes, which can be generated by primordial tensor perturbations produced during Inflation. The detection of such primordial B-modes would provide compelling evidence for the quantum origin of cosmological fluctuations and offer a direct probe of the energy scale of Inflation. However, measuring B-mode polarisation remains extremely challenging. The expected signal is several orders of magnitude weaker than the temperature anisotropies and is significantly contaminated by astrophysical foregrounds, such as Galactic dust and synchrotron emission, as well as by instrumental systematics. Achieving the required sensitivity and control over these systematics therefore necessitates innovative experimental strategies.

The Q&U Bolometric Interferometer for Cosmology (QUBIC) has been designed to address these challenges by combining the high sensitivity of bolometric detectors with the intrinsic systematic control of interferometric techniques. Unlike traditional imaging experiments, QUBIC directly measures the interference pattern produced by an array of horns. This approach introduces a natural redundancy in the instrument response, which can be exploited through a self-calibration procedure to mitigate instrumental systematics.

In this chapter, we present the QUBIC experiment in detail. We first describe the instrumental concept and its key components, with particular emphasis on the principles of bolometric interferometry. We then discuss the scientific objectives of QUBIC. Finally, we present the main instrumental subsystems of the telescope, along with their respective roles and limitations.

The QUBIC collaboration published a series of eight papers in 2022 describing the instrument in detail. These articles cover the overall project [1], its unique spectral capabilities [2], and comprehensive descriptions of its key instrumental components [3, 4, 5, 6, 7, 8]. They form the foundation of this chapter.

1.1 Telescopes for CMB physics

1.1.1 Classical approach: Imager

Before introducing QUBIC instrument, it is interesting to first present the design of numerous modern CMB instruments, called *direct imaging*. Imagers are long-exposure telescope, directly forming images of the sky on their focal planes, where detectors are located. This design is widely adopted by CMB experiments, such as Planck [9], Bicep2/Keck Array [10] or Simons Observatory [11]. These experiments use thousands of detectors, each observing a small patch of the sky. These detectors obserbe photons, measuring their total power. CMB polarisation observations need very high sensitivity detectors. The dominant technology is bolometer, which are equiping nearly every CMB instrument[1].

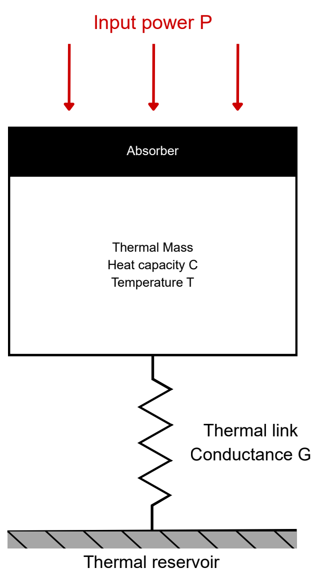

Figure 1.1: Bolometer.

[1]Another type of antenna called Kinetic Inductance Detectors (KIDs) are at the center of many R&D projects, in order to replace bolometers in CMB experiment [12, 13]. KIDs are currently used in NIKA2 camera [14] and tolTEC camera [15]

1.1.2 Innovative design: Bolometric Interferometry

1.2 QUBIC Experiment

1.2.1 QUBIC Collaboration

1.2.2 Observing Site

1.2.3 Science Objectives

1.3 Instrumental Design

1.3.1 Cryogenics

1.3.2 Filters

1.3.3 Half Wave Plate & Polariser

1.3.4 Horns Array

1.3.5 Optical Combiner

1.3.6 Bolometers

Bibliography

[1] J.-Ch. Hamilton et al. “QUBIC I: Overview and science program”. In: *Journal of Cosmology and Astroparticle Physics* 2022.04 (Apr. 2022), p. 034. ISSN: 1475-7516. DOI: 10.1088/1475-7516/2022/04/034. URL: http://dx.doi.org/10.1088/1475-7516/2022/04/034.

[2] L. Mousset et al. “QUBIC II: Spectral polarimetry with bolometric interferometry”. In: *Journal of Cosmology and Astroparticle Physics* 2022.04 (Apr. 2022), p. 035. ISSN: 1475-7516. DOI: 10.1088/1475-7516/2022/04/035. URL: http://dx.doi.org/10.1088/1475-7516/2022/04/035.

[3] S.A. Torchinsky et al. “QUBIC III: Laboratory characterization”. In: *Journal of Cosmology and Astroparticle Physics* 2022.04 (Apr. 2022), p. 036. ISSN: 1475-7516. DOI: 10.1088/1475-7516/2022/04/036. URL: http://dx.doi.org/10.1088/1475-7516/2022/04/036.

[4] M. Piat et al. “QUBIC IV: Performance of TES bolometers and readout electronics”. In: *Journal of Cosmology and Astroparticle Physics* 2022.04 (Apr. 2022), p. 037. ISSN: 1475-7516. DOI: 10.1088/1475-7516/2022/04/037. URL: http://dx.doi.org/10.1088/1475-7516/2022/04/037.

[5] S. Masi et al. “QUBIC V: Cryogenic system design and performance”. In: *Journal of Cosmology and Astroparticle Physics* 2022.04 (Apr. 2022), p. 038. ISSN: 1475-7516. DOI: 10.1088/1475-7516/2022/04/038. URL: http://dx.doi.org/10.1088/1475-7516/2022/04/038.

[6] G. D’Alessandro et al. “QUBIC VI: Cryogenic half wave plate rotator, design and performance”. In: *Journal of Cosmology and Astroparticle Physics* 2022.04 (Apr. 2022), p. 039. ISSN: 1475-7516. DOI: 10.1088/1475-7516/2022/04/039. URL: http://dx.doi.org/10.1088/1475-7516/2022/04/039.

[7] F. Cavaliere et al. "QUBIC VII: The feedhorn-switch system of the technological demonstrator". In: *Journal of Cosmology and Astroparticle Physics* 2022.04 (Apr. 2022), p. 040. ISSN: 1475-7516. DOI: 10.1088/1475-7516/2022/04/040. URL: http://dx.doi.org/10.1088/1475-7516/2022/04/040.

[8] C. O'Sullivan et al. "QUBIC VIII: Optical design and performance". In: *Journal of Cosmology and Astroparticle Physics* 2022.04 (Apr. 2022), p. 041. ISSN: 1475-7516. DOI: 10.1088/1475-7516/2022/04/041. URL: http://dx.doi.org/10.1088/1475-7516/2022/04/041.

[9] J. A. Tauber et al. "Planck pre-launch status: The Planck mission". In: *Astronomy and Astrophysics* 520, A1 (Sept. 2010), A1. DOI: 10.1051/0004-6361/200912983. URL: https://ui.adsabs.harvard.edu/abs/2010A&A...520A...1T.

[10] BICEP2 Collaboration et al. "BICEP2. II. Experiment and three-year Data Set". In: *Astrophysical Journal* 792.1, 62 (Sept. 2014), p. 62. DOI: 10.1088/0004-637X/792/1/62. arXiv: 1403.4302 [astro-ph.CO]. URL: https://ui.adsabs.harvard.edu/abs/2014ApJ...792...62B.

[11] Peter Ade et al. "The Simons Observatory: science goals and forecasts". In: *Journal of Cosmology and Astroparticle Physics* 2019.02 (Feb. 2019), pp. 056–056. ISSN: 1475-7516. DOI: 10.1088/1475-7516/2019/02/056. URL: http://dx.doi.org/10.1088/1475-7516/2019/02/056.

[12] Peter K. Day et al. "A broadband superconducting detector suitable for use in large arrays". In: *Nature* 425.6960 (Oct. 2003), pp. 817–821. DOI: 10.1038/nature02037. URL: https://ui.adsabs.harvard.edu/abs/2003Natur.425..817D.

[13] Jonas Zmuidzinas. "Superconducting Microresonators: Physics and Applications". In: *Annual Review of Condensed Matter Physics* 3 (Feb. 2012), pp. 169–214. DOI: 10.1146/annurev-conmatphys-020911-125022.

[14] R. Adam et al. "The NIKA2 large-field-of-view millimetre continuum camera for the 30 m IRAM telescope". In: *Astronomy & Astrophysics* 609 (Jan. 2018), A115. ISSN: 1432-0746. DOI: 10.1051/0004-6361/201731503. URL: http://dx.doi.org/10.1051/0004-6361/201731503.

[15] Sean A. Bryan et al. "Optical design of the TolTEC millimeter-wave camera". In: *Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy IX*. Ed. by Jonas Zmuidzinas and Jian-Rong Gao. SPIE, 2018, p. 17. DOI: `10.1117/12.2314130`. URL: `http://dx.doi.org/10.1117/12.2314130`.